KAYA Frame Grabbers PoCXP

Application Notes

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KAYA INSTRUMENTS

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Revision History

Version	Date	Notes
2026.1.3	10.2026	Initial release

Table 1 – Revision History

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2 Introduction

2.1 Safety precautions

Please take the time to read through the precautions listed below in order to prevent preventable and unnecessary injuries and damage to you, other personnel or property. Read these safety instructions carefully prior to your first use of the product, as these precautions contain safety instructions that must be observed. Be sure to follow this manual in order to prevent misuse of product.

⚠ Caution! Read Carefully and do not disregard these instructions.
In the event of a failure, disconnect the power supply Disconnect the power supply immediately and contact our sales personnel for repair. Continuing to use the product in this state may result in a fire or electric shock.
If an unpleasant smell or smoking occurs, disconnect the power supply. Disconnect the power supply immediately! Continuing to use the product in this state may result in a fire or electric shock. After verifying that no smoking is observed, contact our sales personnel for repair.
Do not disassemble, repair or modify the product. This may result in a fire or electric shock due to a circuit shortage or heat generation. Contact our sales personnel prior to inspection, modification or repair.
Do not place the product on unstable surfaces. Otherwise, it may drop or fall, resulting in injury to persons or the camera.
Do not use the product if dropped or damaged. Otherwise, a fire or electric shock may occur.
Do not touch the product with metallic objects. Otherwise, a fire or electric shock may occur.
Do not place the product in dusty or humid environments, nor where water may splash. Otherwise, a fire or electric shock may occur.
Do not wet the product or touch it with wet hands. Otherwise, the product may fail or it may cause a fire, smoking or electric shock.
Do not touch the gold-plated sections of the connectors on the product. Otherwise, the surface of the connector may be contaminated by sweat or skin-oil, resulting in contact failure of a connector, malfunction, fire or electric shock due to static electricity discharge.
Do not use or place the product in the following locations. ▪ Unventilated areas such as closets or bookshelves. ▪ Near oils, smoke or steam. ▪ Next to heat sources. ▪ A closed (and not running) car where the temperature becomes high. ▪ Static electricity replete locations ▪ Near water or chemicals. Otherwise, a fire, electric shock, accident or deformation may occur due to a short circuit or heat generation.
Do not place heavy objects on the product. Otherwise, the product may be damaged.
Be sure to discharge static electricity from body before touching any sensitive electronic components. The electronic circuits in the computer and the circuits on the camera are sensitive to static electricity and surges. Improper handling may seriously damage the circuits. In addition, do not let your clothing come in contact with the circuit boards or components. Otherwise, the product may be damaged.

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2.2 Disclaimer

This product should only be used for image capturing and processing. KAYA Vision assumes no responsibility for any damage that may ensue by using the camera for any purpose other than intended, as previously stated. Without detracting from what was previously written, please be advised that the company takes no responsibility for any damages caused by:

- Earthquake, thunder strike, natural disasters, a fire caused by usage beyond our control, willful and/or accidental misuse and/or use under other abnormal and/or unreasonable conditions
- Secondary damages caused by the use of this product or its unusable state (business interruption or others)
- Use of this product in any manner that contradicts this manual or malfunctions due to connection to other devices.
- Damage to this product that is out of our control or failure due to modification.
- Accidents and/or third parties that may be involved.

Additionally, KAYA Vision assumes no responsibility or liability for:

- Erasure or corruption of data caused by the use of this product
- Any consequences or other abnormalities following the use of this product

Repairs to this product are carried out by replacing it on a chargeable basis and not by repairing the faulty device. Non-chargeable replacement is offered for the initial failure, as long as it is reported no later than two weeks post-delivery of the product.

Disabling automatic PoCXP management, either system-wide or for an individual grabber, may damage the frame grabber or connected camera. The user is responsible for equipment damage resulting from operation with automatic management disabled.

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3 Overview

Power over CoaXPress (PoCXP) supplies power to a compatible camera through its CoaXPress connection. This note explains how to configure automatic PoCXP management and control power on individual frame grabber links in KAYA Vision Studio.

RECOMMENDED CONFIGURATION:
Keep automatic PoCXP management enabled for the system and automatic monitoring enabled for each supported frame grabber. Let the monitor control power according to the link connection and the presence of a camera.

3.1 PoCXP control scopes

The system preference determines startup behavior. PoCXPAutoActive enables monitoring for one grabber. The connection selector and commands determine which links are automatically managed or forced off.

3.1.1 System startup setting

The Automatic PoCXP management preference applies to the entire system and determines whether monitoring starts at system startup.

3.1.2 Frame grabber monitoring

PoCXPAutoActive enables or disables automatic monitoring for the selected grabber at runtime.

3.1.3 Physical link control

CxpPoCxpHostConnectionSelector chooses one physical link or all links on the selected grabber. CxpPoCxpAuto restores automatic control and CxpPoCxpTurnOff forces the selected links off.

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3.2 Hardware support

Automatic monitoring requires compatible hardware, firmware, and software. The table summarizes the support information for the frame grabber families covered by this note. Confirm availability on the installed setup using PoCXPAutoAvailable.

Frame grabber	Firmware	Automatic monitoring
Komodo II CoaXPress	All versions	Supported
Komodo III CoaXPress	All versions	Supported
Predator II CoaXPress	All versions	Supported

Table 2 – Automatic PoCXP monitoring support

4 Implementation

4.1 System preferences

Open Preferences > Advanced. The Automatic PoCXP management setting controls whether automatic monitoring starts for the entire system. Changes to this setting require a system reboot.

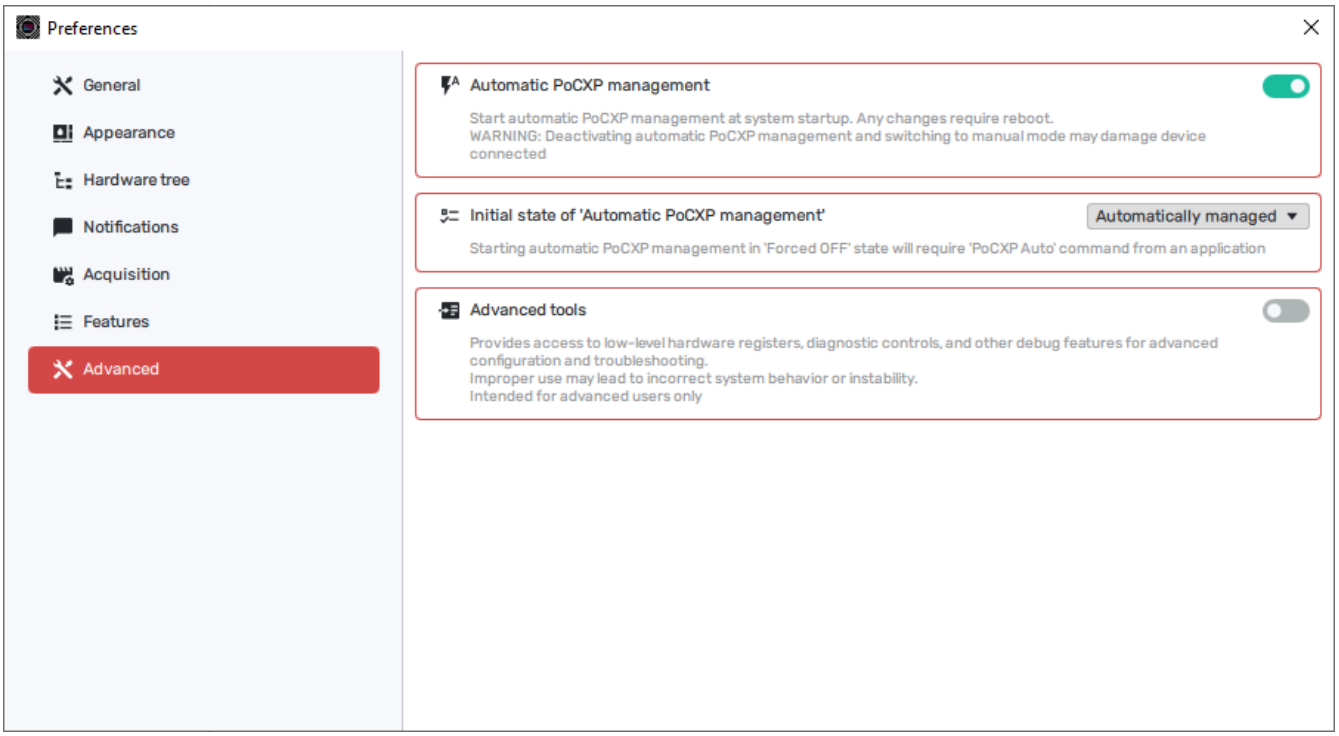


Figure 1 – System preferences for automatic PoCXP management

1. Enable Automatic PoCXP management.
2. Set Initial state of 'Automatic PoCXP management' to Automatically managed for normal operation.
3. Reboot the system after changing Automatic PoCXP management so that the startup setting takes effect.

WARNING:
Disabling automatic PoCXP management allows manual control and may damage the frame grabber or a connected camera. The user is responsible for any equipment damage resulting from operation with automatic PoCXP management disabled.

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4.2 Initial state of automatic management

If Initial state of 'Automatic PoCXP management' is set to Forced Off, automatic management starts with link power forced Off. Select the required link or All and execute CxpPoCxpAuto to allow automatic power control. Keeping monitoring enabled alone does not override a link that has been forced off.

4.3 Automatic monitoring for a frame grabber

Select the required frame grabber in KAYA Vision Studio and open its feature list. Expand Device Control > PoCXP Control. The figure below shows the monitoring switch, link power indicators, connection selector, and PoCXP commands. The available links depend on the frame grabber model.

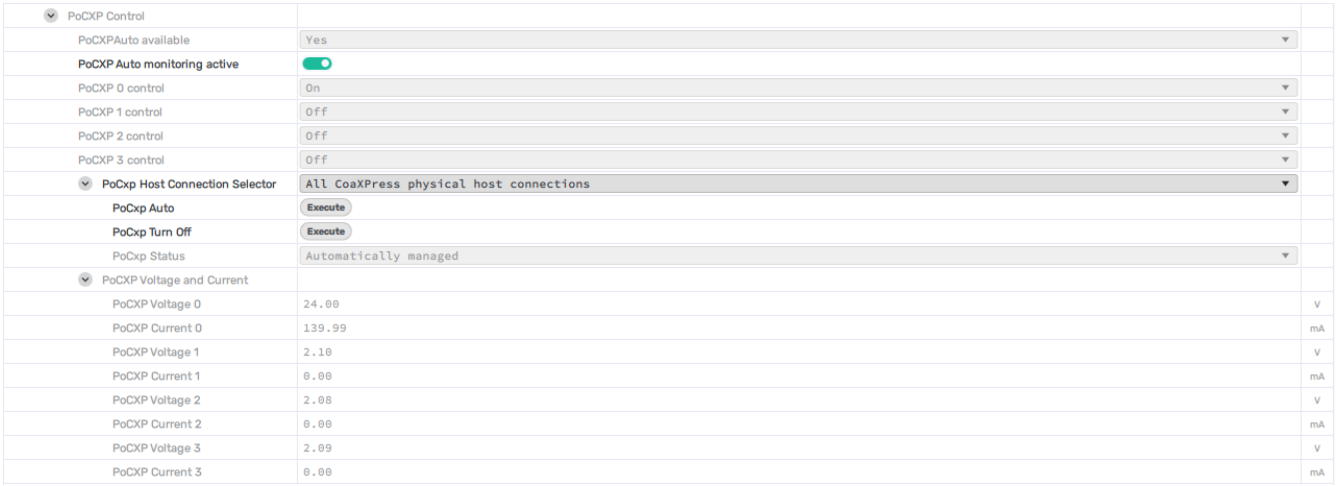


Figure 2 – Frame grabber features for PoCXP control and monitoring

1. Check automatic monitoring support

Read PoCXPAutoAvailable, displayed as PoCXPAuto available. A value of Yes indicates that the current hardware, firmware, and software combination supports automatic PoCXP monitoring. Check this feature on the actual frame grabber before relying on automatic operation.

2. Enable monitoring for the selected grabber

PoCXPAutoActive is displayed as PoCXP Auto monitoring active. Enable it to let the monitor manage PoCXP for this frame grabber according to the link connection and camera presence. Enabled is the recommended state. This setting applies to the selected grabber at runtime; it does not replace the system startup preference.
Disabling PoCXPAutoActive stops automatic monitoring for this grabber and allows manual control through the available PoCXP link features. To force selected links off while retaining automatic monitoring for the grabber, use CxpPoCxpTurnOff as described on the next page.

WARNING:
Disabling PoCXPAutoActive may damage the frame grabber or a connected camera. The user is responsible for any equipment damage resulting from operation with automatic monitoring disabled.

4.4 Control one link or all links

Use the Gen<I>Cam SFNC PoCXP commands CxpPoCxpAuto and CxpPoCxpTurnOff together with CxpPoCxpHostConnectionSelector to control the required physical host connections. Keep PoCXPAutoActive enabled when using this workflow.

4.4.1 Select the connections

CxpPoCxpHostConnectionSelector is displayed as PoCXP Host Connection Selector. Select one physical host connection, such as CoaXPress physical host connection 0, or select All CoaXPress physical host connections. All applies to every available physical link on the selected grabber, not to other grabbers in the system.

The selector only chooses the targets for the next command and for the status readout. Changing it does not change link power. For feature access in an application, the selector entries are All and Link0 through LinkN, according to the available physical links.

4.4.2 Return the selected links to automatic control

1. Choose the required connection or All in PoCXP Host Connection Selector.
2. Click Execute beside PoCXP Auto (CxpPoCxpAuto).
3. Read PoCXP Status (CxpPoCxpStatus) and the corresponding PoCXP link indicators.

CxpPoCxpAuto places the selected connections under automatic power control. The monitor decides when to enable or disable PoCXP according to the connection and camera presence. Command completion does not mean that power has already been enabled or that the camera has finished starting.

4.4.3 Turn the selected links off

1. Choose the required connection or All in PoCXP Host Connection Selector.
2. Click Execute beside PoCXP Turn Off (CxpPoCxpTurnOff).
3. Verify Forced Off in PoCXP Status and Off in the corresponding PoCXP link indicators. To resume automatic control, select the same connection or connections and execute PoCXP Auto.

CxpPoCxpTurnOff forces PoCXP off on the selected connections. It does not disable PoCXPAutoActive for the entire grabber. Selecting one link leaves the other links under their existing control settings.

4.4.4 Read the selected connection status

PoCXP Status	Meaning
Automatically managed	The selected connections are under automatic control; their power may currently be On or Off.
Forced Off	PoCXP is forced off for the selected connections.
Mixed statuses	When All is selected, the connections do not share one status. Select links individually to inspect them.
Tripped	PoCXP has tripped on the selected connection. Check the connected equipment and fault condition before restoring power.

Table 3 – PoCXP connection status

4.5 Read link power and allow for startup time

PoCXP0 through PoCXPN correspond to the physical links on the selected frame grabber. The index starts at 0, so N is the number of physical links minus one. For example, a four-link grabber exposes PoCXP0 through PoCXP3. While automatic monitoring is enabled, these features are read-only. Use them as indicators of whether PoCXP is enabled on each link. They remain tied to their own link numbers regardless of the connection selector setting.

Feature value	Interpretation
On (PoCXPOn)	PoCXP is enabled on that physical link.
Off (PoCXPOff)	PoCXP is disabled on that physical link.

Table 4 – Per-link PoCXP power indications

In Figure 2, PoCXP0 is On and PoCXP1 through PoCXP3 are Off. PoCXP Status is Automatically managed, illustrating that automatic control and actual link power state are separate indications.

4.5.1 Allow time for power to become available

Automatic PoCXP monitoring may take time to enable power on a grabber link after a camera is connected or automatic control is resumed. Account for this interval in application timing or monitor the corresponding PoCXP0 through PoCXPN features until the required links report On.

An On value confirms that PoCXP is enabled at the frame grabber link. The camera may still need additional time to complete its own startup before it can be used. Allow for that startup time as well and handle a timeout if the expected link power state is not reached.

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4.6 Manual control when monitoring is disabled

Manual operation is available when automatic monitoring is disabled, or when the hardware does not support it. The recommended operating mode is automatic monitoring whenever it is supported.

WARNING:
Operating with automatic PoCXP management or PoCXPAutoActive disabled may damage the frame grabber or a connected camera. The user is responsible for any resulting equipment damage. Do not connect or disconnect a camera while PoCXP is manually enabled; turn off power on the affected links first.

1. Select the required grabber and confirm that automatic monitoring is disabled for it. Changing the system startup preference requires a reboot; PoCXPAutoActive controls the selected grabber at runtime.
2. Use the writable PoCXP 0 control through PoCXP N control features to set the required links to On or Off. Change only the links needed for the connected equipment.
3. When manual operation is no longer required, restore the system startup preference and enable PoCXPAutoActive on supported grabbers. Select the required links and execute CxpPoCxpAuto to return them to automatic control.

The SFNC selector and command controls may be unavailable while automatic monitoring is disabled. Use the writable per-link PoCXP features for manual control and restore monitoring before using the automatic-control command workflow.

4.7 Further reading

The CxpPoCxpAuto and CxpPoCxpTurnOff command definitions follow the Gen<I>Cam Standard Features Naming Convention. KAYA Vision Studio applies these commands to the physical connections chosen by CxpPoCxpHostConnectionSelector.

EMVA Gen<I>Cam Standard Features Naming Convention 2.8

REFERENCES

Supported vision standards:

CoaXPress GEN<i>CAM

Vision Point documentation:

Vision Point

TECHNICAL SUPPORT AND PROFESSIONAL SERVICE

If you searched the documents and could not find the answers you need, contact KAYA Instruments support service:

- Create a support request on the web: support.kayainstruments.com
- Our knowledge base is available on: kb.kayainstruments.com

Visit us at kayainstruments.com for comprehensive information.

SUBMITTING A SUPPORT REQUEST

When opening a support request, please provide the following information when applicable:

For Frame Grabbers:	For Camera:	For Range Extender:
• Vision Point Diagnostic Info* • Serial number of Frame Grabber • Camera model • SFP+ module model • CoaXPress/Fiber cable model and length • External power or PoCXP • PC motherboard model *In the Vision Point app, use menu option Help > Collect diagnostic info.	• Vision Point Diagnostic Info (or frame grabber being utilized) • Serial Number of Camera • XML dump and/or description of how the camera is being utilized • Description of issue • SFP+ module model • CoaXPress/Fiber cable model and length • External power or PoCXP	• Range Extender Model • Serial Number of Range Extender • SFP+ module model • CoaXPress/Fiber Cable model and length • PC configuration • Operating System • Software Revision • Camera and Frame Grabber Manufacturer and Model

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